Natalia Isabel Martínez Gómez

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Physics student at the Universidad del Quindío (Armenia, Colombia), interested in particle physics, astroparticle physics and astronomy. For my thesis I study the Standard Model and neutrino physics, carrying out a statistical analysis of the first KATRIN science run to estimate the neutrino mass and explore the existence of sterile neutrinos.

Throughout my training I have focused on scientific data analysis and processing: statistical fitting, machine learning, astronomical data and numerical simulation of quantum and dynamical systems, mainly in Python and Fortran.

PROJECTS

Analysis of the tritium beta-decay spectrum in KATRIN to set limits on sterile-neutrino parameters · Thesis (in progress)
Advisor: Víctor Saúl Basto González, Ph.D. · Universidad del Quindío
Statistical analysis of the public data from the first KATRIN science run (KNM1) to test the eV-scale sterile-neutrino hypothesis suggested by the reactor and gallium anomalies, independently of reactor fluxes.

- Theoretical background: massive neutrinos beyond the Standard Model, lepton mixing and the 3+1 scheme; the tritium beta spectrum from Fermi theory, including the Fermi function, the molecular final-state distribution of T_2 and the KATRIN MAC-E filter.
- Fortran 2008 code developed together with my advisor: χ^2 fit with a pull term and profiling of nuisance parameters. Result: $m_\beta{}^2 = -1.15 \pm 0.81$ eV^2 (68% CL), consistent with the value published by KATRIN.
- Exclusion contours in the ($\sin^2 2\theta_{ee}$, $\Delta m^2{}_{41}$) plane for the 3+1 scenario, with confidence levels given by Wilks' theorem, compared with the results of Giunti, Li and Zhang (2019).

EDUCATION

Physics · 2020 – present
Universidad del Quindío, Armenia · All coursework completed; thesis in progress.

Soil Laboratory Technician · 2018 – 2020
SENA Nariño, Pasto

EXPERIENCE

Teaching Assistant, Physics Laboratories · 08/2023 – 05/2025
Universidad del Quindío, Armenia

- Supported students during lab sessions: proper use of the equipment and choice of the right materials for each experiment.
- Set up experiments ranging from classical mechanics to modern physics.
- Answered students' questions during the lab sessions.
- Guided tours of the laboratories for high-school students, explaining the experiments.

CONFERENCES

XVI Latin American Symposium on High Energy Physics (SILAFAE 2026) · Nov 30 – Dec 4, 2026
Universidad de los Andes, Bogotá
Poster (accepted): *"An independent reanalysis of the KATRIN bound on 3+1 active–sterile neutrino mixing"*

XXX Congreso Nacional de Física (Colombian National Physics Congress) · Oct 8–10, 2025
Universidad Nacional de Colombia, Manizales
Poster: *"Analysis of the tritium beta-decay spectrum in KATRIN to set limits on sterile-neutrino parameters"*

9th Colombian Meeting on High Energy Physics (ComHEP) · Dec 2–6, 2024
Universidad de Nariño, Pasto · Attendee

XXIX Congreso Nacional de Física · Sep 21–23, 2022
With the Sociedad Colombiana de Física · Universidad del Quindío, Armenia · Attendee

SCHOLARSHIPS

Generación E · Colombian Ministry of National Education
Scholarship that funded my undergraduate studies in Physics.

Sports scholarship · Universidad del Quindío
Member of the university's women's futsal team.

TOOLS

Programming: Python · Fortran

Data analysis and processing: NumPy · SciPy · pandas · scikit-learn · Astropy

Simulation and visualization: QuTiP · Matplotlib

Workflow: Jupyter · Git

LANGUAGES

Spanish (native) · English (B1)